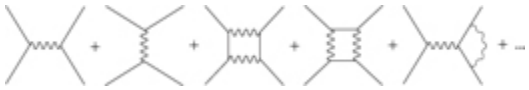
impossible energies, and tossing such particles back and forth between themselves. A real photon has exactly zero mass, but the mass of a virtual photon can be absolutely anything. What we mean by “virtual particles” are subtle distortions in the wave function of a collection of quantum fields. Sometimes they are called “fluctuations” or simply “modes” (referring to a vibration in a field with a particular wavelength). But everyone calls them particles, and they can be successfully represented as lines within Feynman diagrams, so we can call them that.

The diagram we drew for an electron and a positron scattering off each other isn’t the only one we could possibly draw; in fact, it’s just one of an infinite number. The rules of the game tell us that we should sum up all of the possible diagrams with the same incoming and outgoing particles. We can list such diagrams in order of increasing complexity, with subsequent diagrams containing more and more virtual particles.



The final number we obtain is an amplitude, so we square it to get the probability of such a process happening. Using Feynman diagrams, we can calculate the probability of two particles scattering off each other, of one particle decaying into several, or for particles turning into other kinds of particles.

An obvious worry pops up: If there are an infinite number of diagrams, how can you add them all up and get a sensible result? The answer is that diagrams contribute smaller and smaller amounts as they become more complicated. Even though there are an infinite number of them, the sum total of all the very complicated ones can be a tiny number. In practice, as a matter of fact, we often get quite accurate answers by calculating only the first few diagrams in the infinite series.

There is one subtlety along the way to this nice result, however. Consider a diagram that has a loop in it—that is, where we can trace around some set of particle lines to form a closed circle. Here is an electron and a positron exchanging two photons: